

Adarsh Ravi Rao

Software Engineer | Java | Spring Boot | Kafka | Kubernetes | LLM Integration

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Summary

Java backend software engineer with 5+ years of experience building **Spring Boot microservices**, **Kafka event-driven systems**, and **Kubernetes-deployed platforms** across Uber, Dell Technologies, and Capgemini. Delivers measurable outcomes: reduced deployment cycles from **3 days to 18 hours**, migrated **850,000+ records** to a microservices architecture with **zero data loss**, and diagnosed **65+ production issues** that prevented **200+ hours of downtime**. Also builds production **LLM integrations** on top of that backend foundation, including a **LangChain document processing pipeline** and a **LangGraph conversational AI backend**. MS in Computer & Information Sciences, University of Texas at Arlington.

Professional Experience

Software Developer

June 2026 – Present

Uber

USA

- Engineered **Java 17**, **Spring Boot** microservice components and distributed **REST APIs**; integrated internal observability tooling (**M3**, **Jaeger**) and optimized **DocStore/PostgreSQL** queries to reduce daily sync time across global platforms.
- Configured automated **CI/CD** pipelines within the engineering monorepo using **Bazel** and internal deployment tooling (**Up CD**), embedding security scans and rollback logic to reduce release cycles by **3.5 hours**.
- Built backend orchestration for an internal **LLM-powered document parser** using **Java** and **LangChain**, processing an initial rollout of **12,000+ documents** as part of an effort projected to reduce quarterly review time by **11 days**.
- Contributed to infrastructure planning and **Kubernetes** migration efforts using **Helm**, resolving high-priority technical debt and improving system reliability across 2 engineering units.

Software Engineer

May 2025 – Aug 2025

BotStacks

USA

- Built an initial **conversational AI backend** prototype using **Node.js**, **LangGraph**, and **OpenAI APIs**, establishing the core **RESTful** architecture behind a customer-facing **React.js** chatbot interface.
- Automated containerized deployments using **Docker**, **GitHub Actions**, and **AWS** (ECS/S3) to establish initial **CI/CD** pipelines, accelerating the transition from local development to cloud staging environments.
- Improved **LLM query processing speed by 30%** during early integration testing by incorporating **Redis** caching to optimize AI context retrieval.
- Partnered with frontend engineering teams to define **API contracts** and manage state across user sessions, ensuring reliable data flow between the AI orchestration layer and the UI.
- Designed a **session-aware orchestration layer** using **LangGraph** and **Node.js** to route multi-turn chatbot requests, reducing dropped conversations during early staging tests.
- Implemented **secure API gateways** for **OpenAI** traffic with request throttling and retry logic on **AWS ECS**, improving backend stability for the **React.js** chatbot rollout.

Software Engineer

May 2023 – Jan 2024

Dell Technologies

India

- Engineered **Java 17 Spring Boot** microservices to manage artifact ownership, ensuring data sovereignty for 12 enterprise applications and generating **4,000+ verifiable audit logs**.
- Implemented an **Apache Kafka** event-driven architecture to automate ownership updates, removing 12 manual intervention points per release and decreasing support ticket volume by **45 tickets monthly**.
- Strengthened **API security** using **JWT** and **Spring Security**, enabling 5 internal governance platforms hosted on **Microsoft Azure** to sync risk data while accelerating deployment cycles from **3 days to 18 hours** via **Jenkins**.
- Optimized data retrieval through **SQL query tuning** and **React.js** code-splitting, reducing portal load times by **2.5 seconds** for a user base of **30,000 monthly active users**.
- Standardized **10+ internal dashboards** using **TypeScript** and shared UI components, resolving 25 recurring UI inconsistencies per sprint and accelerating cross-functional developer handoff.

Software Engineer

Apr 2021 – May 2023

Capgemini

India

- Built event-driven **Spring Boot** microservices using **Kafka** for asynchronous processing, enabling horizontal scalability for **\$2.2M+ in monthly transaction volume** while maintaining high availability through fault-tolerant design patterns.
- Refined data persistence layers using **Spring Data JPA** and **Hibernate**, integrating **25+ RESTful endpoints** with **Oracle Database** and reducing query latency by **300ms** through indexed execution plans.
- Orchestrated containerized deployments using **Docker** and **Jenkins** on **Azure Kubernetes Service (AKS)**, managing 4 staging environments and facilitating 3 zero-downtime production releases per month within an **Agile** framework.
- Diagnosed **65+ high-priority production bottlenecks** using **Dynatrace** and **Log Analytics**, preventing **200+ hours of cumulative downtime** and improving service reliability across the procurement lifecycle.

Junior Software Developer

Jun 2019 – Mar 2021

Robosoft Technologies

India

- Programmed web modules using **JavaScript**, **jQuery**, and **Bootstrap**, delivering 5 client-facing applications that passed accessibility audits for US-based end users.
- Built **SQL Server** database schemas and **PL/SQL** stored procedures to support internal CRM tools, managing an enterprise dataset of **10M+ customer interaction records**.
- Translated **Figma** wireframes into high-fidelity **HTML5** and **SASS** components using **Angular**, reducing design-to-code friction across 4 major projects.
- Authored **150+ unit tests** using **JUnit** and **Mockito**, ensuring comprehensive logic validation and identifying 12 critical structural flaws prior to final QA.
- Eliminated 12 manual data entry workflows by developing **Python** automation scripts, saving the operations team **25 hours of manual labor per month**.

Technical Skills

- **Languages & Programming:** Java (8–17), Python, TypeScript, JavaScript (ES6+), SQL, PL/SQL, HTML5, CSS3
- **Frameworks & Libraries:** Spring Boot, Spring MVC, Spring Data JPA, Hibernate, React.js, Angular, Node.js, LangChain, Bootstrap, SASS, jQuery
- **Backend & APIs:** RESTful APIs, Microservices Architecture, API Design, Swagger/OpenAPI, Postman
- **Security & Authentication:** JWT, Spring Security, OAuth 2.0, Role-Based Access Control (RBAC)
- **AI & LLM Systems:** LangGraph, LLM Integration (OpenAI), Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone), Multi-Agent Systems, Prompt Engineering, AI Evaluation & Guardrails
- **Databases & Data Engineering:** PostgreSQL, Oracle Database, Redis, SQL Server, DocStore, Data Modeling, Query Optimization
- **Cloud & DevOps:** Microsoft Azure, Azure Kubernetes Service (AKS), AWS (ECS, S3), Docker, Kubernetes (K8s), Helm, Bazel, Jenkins, GitHub Actions, Git/GitHub
- **Messaging & Distributed Systems:** Apache Kafka, Event-Driven Architecture, Asynchronous Processing, Batch Processing
- **Testing & Quality Engineering:** JUnit, Mockito, Unit Testing, Test-Driven Development (TDD)
- **Monitoring & Observability:** Dynatrace, Splunk, Azure Application Insights, Log Analytics, M3, Jaeger
- **CI/CD & Deployment:** CI/CD Pipelines, Continuous Integration, Continuous Deployment
- **Design & Frontend Engineering:** Figma, Responsive Web Design
- **Methodologies & Practices:** Agile (Scrum), Software Development Life Cycle (SDLC), Microservices Architecture, Code Reviews, Sprint Planning

Projects

AI Reliability & Hallucination Detection System

- Developed an **AI validation layer** to detect hallucinations in **LLM outputs** by integrating **RAG pipelines**, vector similarity scoring (**Pinecone**), and confidence-based filtering, addressing inconsistent responses in enterprise AI workflows.
- Built feedback loops and monitoring pipelines using **Python**, **LangChain**, and **OpenAI APIs**, reducing incorrect responses by **35%** and improving trust in production AI systems.

Multi-Agent Workflow Automation Platform

- Built a scalable **multi-agent system** with **LangGraph**, **Apache Kafka**, and **Node.js** to automate enterprise workflows such as ticket resolution and data processing across distributed services.
- Designed an event-driven pipeline using **Docker**, **Kubernetes**, and **AWS**, enabling concurrent task execution and cutting manual processing time by **60%** while maintaining fault-tolerant scalability.

Education

Master of Science, Computer & Information Sciences

Jan 2024 – Dec 2025

University of Texas at Arlington

Arlington, TX